Genome Coverage and Copy Number Variation Analysis of Human Cytomegalovirus from Clinical Whole Genome Sequencing Data

Jânice Roberta de Paula

2026-03-28

Contents

1 Introduction **2**

2 Materials and Methods **3**

2.1 Dataset 3

2.2 Reference Genome 3

2.3 Data Processing 4

3 Visualization and Genomic Annotation **11**

4 Results **11**

4.1 Variant Distribution Across the Genome 11

4.2 Genes with Highest Mutation Frequency 12

4.3 Mutation Density Normalized by Gene Length 13

4.4 Genome Coverage Depth 13

4.5 Copy Number Variation Analysis 14

5 Discussion **14**

6 Conclusion **14**

Abstract

Human cytomegalovirus (HCMV) is a large double-stranded DNA virus responsible for severe infections in immunocompromised individuals and newborns. Advances in next-generation sequencing (NGS) technologies allow detailed investigation of viral genomic diversity directly from clinical samples.

In this study, whole genome sequencing data from BioProject PRJNA945420 were analyzed to investigate genomic variability and structural variation in HCMV. Sequencing reads generated using both Illumina and Oxford Nanopore platforms were aligned to the HCMV reference genome (strain Merlin). Variant detection was performed and genomic coordinates were integrated with gene annotations using Bioconductor tools in R.

Variant distribution across the genome was evaluated, and genes presenting the highest mutation frequencies were identified. Genome coverage depth was also analyzed to investigate read distribution and detect potential copy number variations (CNVs). CNV profiles were inferred using coverage-based normalization relative to median genome coverage.

The analysis revealed heterogeneous distribution of variants across the viral genome, with specific genes presenting higher mutation frequencies. Coverage-based CNV analysis identified genomic regions with potential duplications and deletions, suggesting structural variability within the viral population.

These results demonstrate the utility of combining NGS data with bioinformatic analysis pipelines to characterize viral genomic variation and structural diversity directly from clinical sequencing data.

1 Introduction

Human cytomegalovirus (HCMV), belonging to the Herpesviridae family, is a double-stranded DNA virus widely distributed in the human population. Although infection is usually asymptomatic in immunocompetent individuals, HCMV can cause serious illness in newborns, immunosuppressed patients, and transplant recipients, being one of the main causes of congenital infections and clinical complications associated with immunosuppression. The high genomic variability of the virus plays an important role in viral adaptation, pathogenicity, and response to antiviral treatments.

With advances in next-generation sequencing (NGS) technologies, it has become possible to investigate the genetic diversity of viruses with greater resolution directly from clinical samples. Approaches combining short-read (Illumina) and long-read (Oxford Nanopore) platforms allow for the reconstruction of viral genomes with greater structural accuracy, as well as enabling the identification of complex genomic variations. This type of hybrid strategy is particularly useful for large genome viruses, such as HCMV, whose genome is approximately 235 kb and has repetitive regions and frequent structural rearrangements.

The BioProject PRJNA945420 contains whole genome sequencing data obtained directly from clinical samples with high viral load, using Illumina and Nanopore technologies. This data was generated with the aim of enabling direct sequencing of HCMV genomes without the need for viral culture, allowing for a more accurate analysis of the viral diversity present in the host. The generated reads were deposited in the Sequence Read Archive (SRA) and used for genomic reconstruction and studies of the virus's structural variability.

Due to the high sequencing depth obtained in these samples, the data also allow for quantitative analyses based on coverage depth, such as the detection of copy number variations (CNVs) along the viral genome. These variations may reflect duplication or deletion events in genomic regions and may be associated with evolutionary processes of the virus, adaptation to the host, or changes in viral gene expression. In this work, a genomic coverage analysis and CNV detection were performed using sequencing data available in BioProject PRJNA945420. The objective was to identify possible regions of the viral genome exhibiting duplications or relative deletions, contributing to the characterization of the genomic architecture of HCMV in clinical samples with high viral load.

2 Materials and Methods

2.1 Dataset

The sequencing data analyzed in this study were obtained from the BioProject PRJNA945420, available in the National Center for Biotechnology Information (NCBI) database. This project contains whole genome sequencing data of human cytomegalovirus (HCMV) obtained directly from clinical samples presenting high viral load.

The dataset includes sequencing reads generated using a hybrid sequencing strategy, combining:
- *Illumina sequencing, producing short high-accuracy reads*
- *Oxford Nanopore sequencing, producing long reads that facilitate the resolution of structurally complex genomic regions.*

This combination enables improved genome reconstruction and facilitates the detection of structural genomic variation.

All raw sequencing reads are available in the Sequence Read Archive (SRA) and were processed for downstream coverage depth and structural variation analyses.

2.2 Reference Genome

Reads were aligned against the reference genome of Human cytomegalovirus strain Merlin, which is widely used as the standard reference genome for HCMV genomic analyses.

Reference genome information:
- *Organism: Human cytomegalovirus*
- *Reference accession: NC_006273*
- *Genome size: ~235 kb*
- *Genome type: double-stranded DNA*

The genome annotation file in GenBank/GFF format was used to identify genomic regions and viral genes during downstream analyses.

2.3 Data Processing

The analysis followed a standard bioinformatics workflow for viral genome analysis.

2.3.1 Read Alignment

Sequencing reads were aligned to the HCMV reference genome using alignment tools appropriate for each sequencing platform.

Short reads generated by Illumina were aligned using a short-read aligner optimized for high-accuracy mapping. Long reads produced by Nanopore sequencing allow better resolution of repetitive and structurally complex genomic regions.

The alignment process generated BAM files, which were subsequently sorted and indexed for downstream analysis.

2.3.2 Variant File Processing

Variant information was stored in VCF (Variant Call Format) files. These files contain genomic coordinates and details of detected variants relative to the reference genome.

The VCF files were imported into R using Bioconductor packages, enabling integration with genomic annotation data.

Genomic ranges corresponding to detected variants were compared with annotated viral genes using overlap-based methods implemented in the GenomicRanges framework.

Variant information was stored in VCF (Variant Call Format) files. These files contain genomic coordinates and details of detected variants relative to the reference genome.

The VCF files were imported into R using Bioconductor packages, enabling integration with genomic annotation data.

Genomic ranges corresponding to detected variants were compared with annotated viral genes using overlap-based methods implemented in the GenomicRanges framework.

2.3.3 Variant Annotation

The functional annotation of the detected variants was performed using SnpEff, a widely used tool for predicting the potential impact of genetic variants on genomic features such as genes,

transcripts, and coding sequences.

SnpEff uses genome annotation files to determine whether variants occur in coding regions, introns, intergenic regions, or regulatory elements, and predicts their potential biological effects.

For this analysis, the HCMV genome annotation file (GFF format) corresponding to the reference genome was used to build the SnpEff database. The variant file (VCF) generated after variant calling was then annotated using SnpEff to identify the genomic context of each mutation.

The annotated VCF file was subsequently imported into R for downstream analysis and visualization.

2.3.4 Variant Analysis and Visualization in R

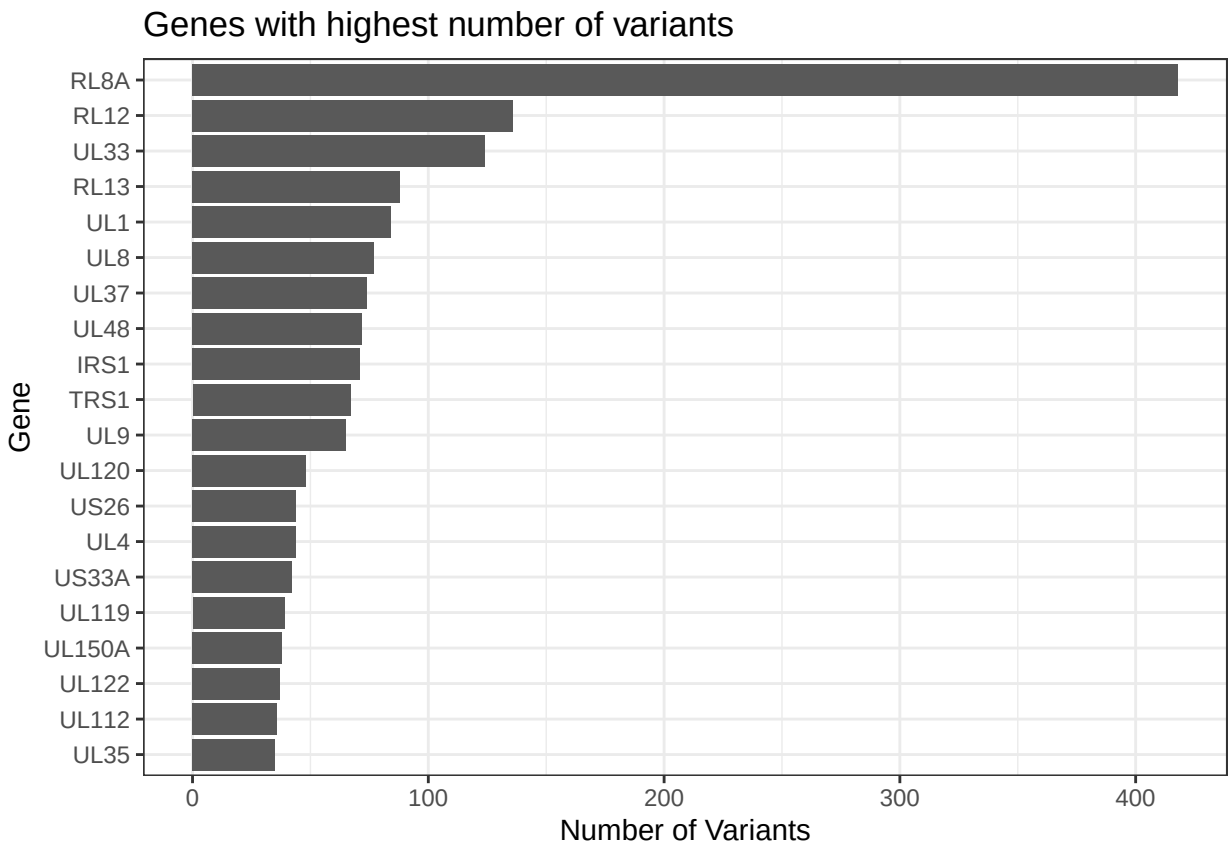
The annotated VCF file was imported into R using the Bioconductor package VariantAnnotation. Genomic annotation data were imported using rtracklayer, enabling integration of variant data with gene coordinates.

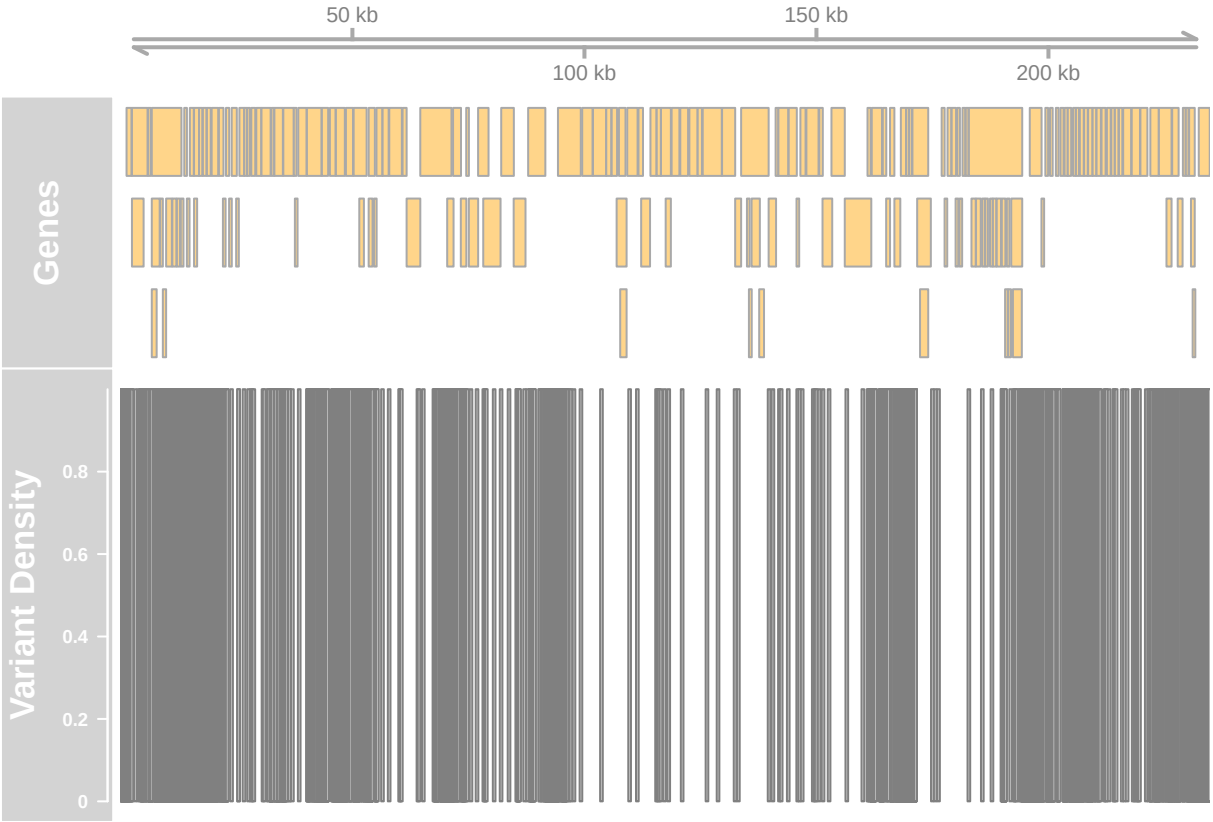
Genomic overlaps between detected variants and annotated genes were identified using functions from the GenomicRanges package.

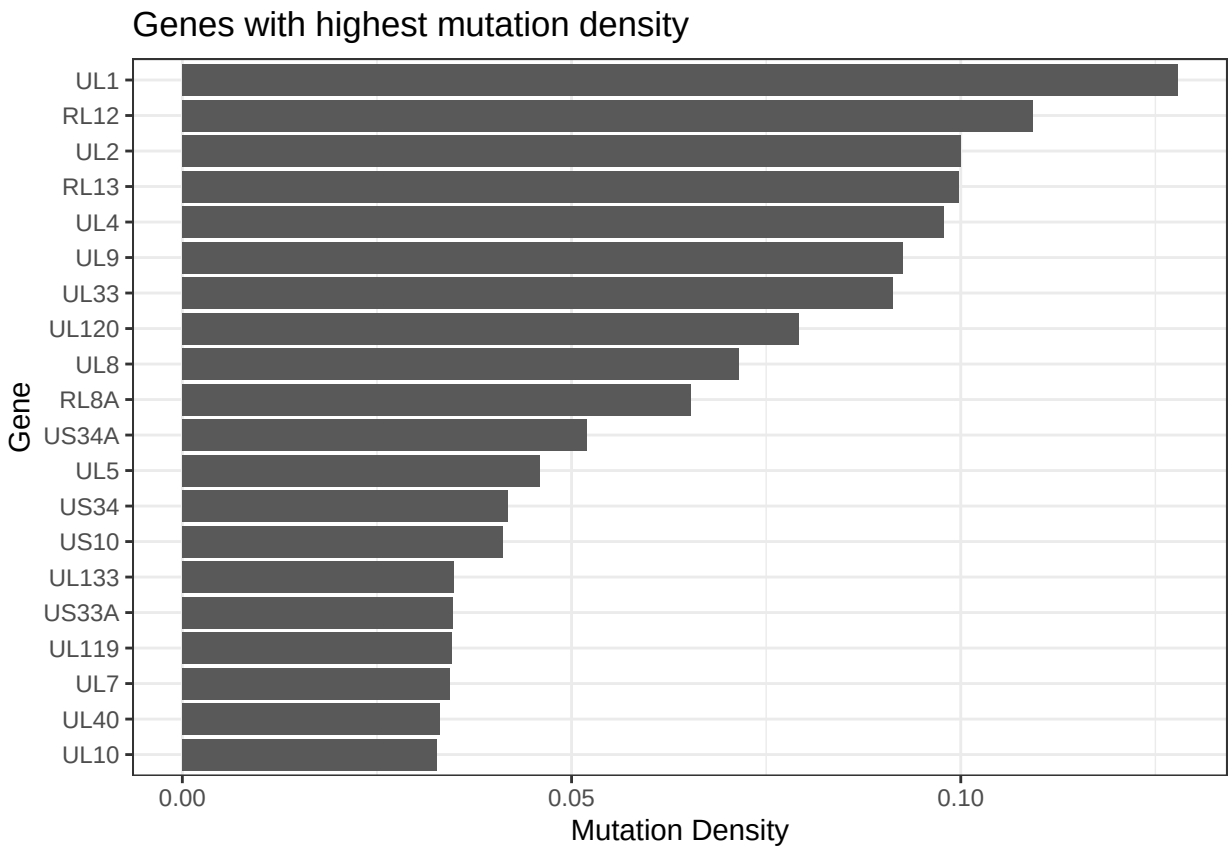
Variants located within genes were extracted and summarized to determine the number of mutations affecting each viral gene. These results were used to identify genes presenting the highest mutation frequency.

Visualization of the results was performed using ggplot2 for statistical plots and Gviz for genome-level representation of gene structures and variant locations.

Additionally, mutation density was calculated by normalizing the number of detected variants by the length of each gene, allowing comparison of mutation rates across genes of different sizes.


Genes with highest number of variants
RL8A
RL12
UL33
RL13
UL1
UL8
UL37
UL48
IRS1
TRS1
UL9
UL120
US26
UL4
US33A
UL119
UL150A
UL122
UL112
UL35
Gene
0
100
200
300
400
Number of Variants


50 kb
150 kb
100 kb
200 kb
Genes
Variant Density
0.8
0.6
0.4
0.2
0



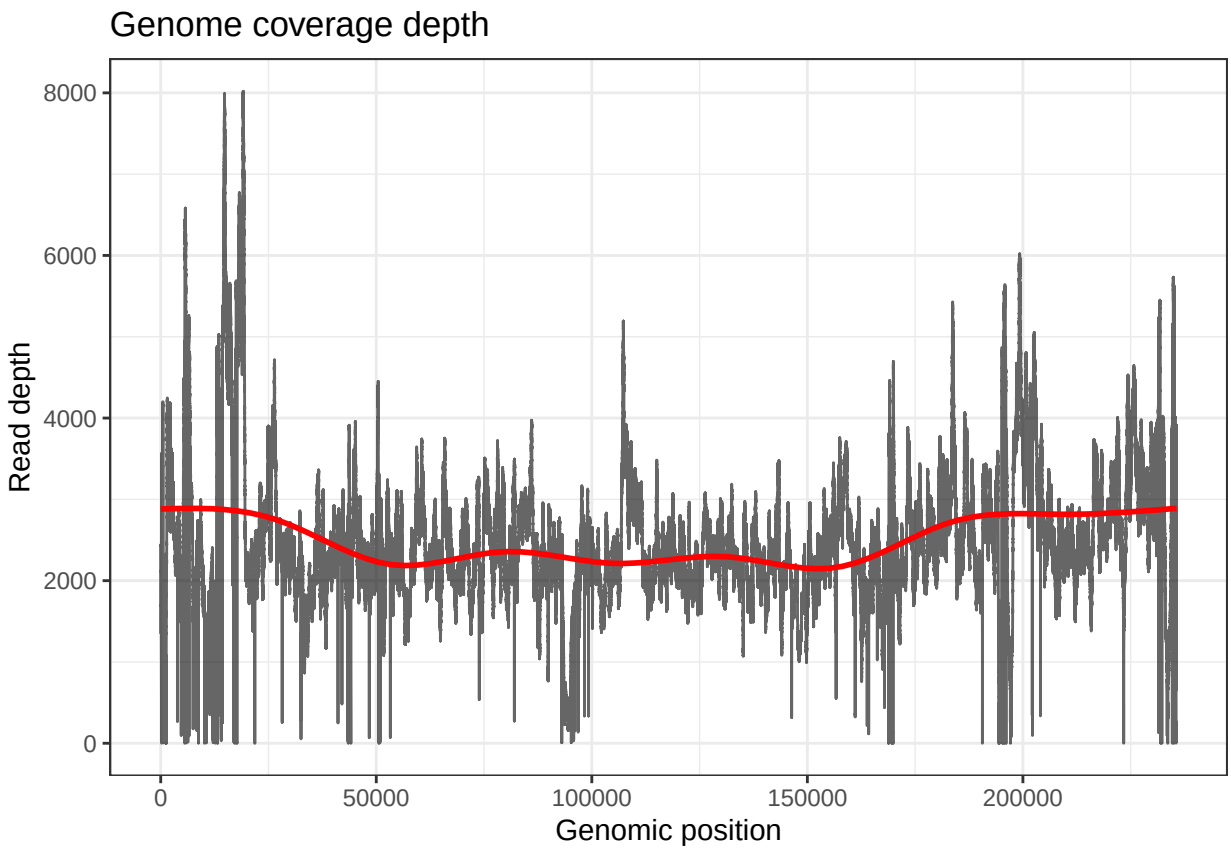
2.3.5 Coverage Depth Calculation

Genome coverage depth was calculated to evaluate the distribution of sequencing reads across the viral genome.

Coverage depth represents the number of sequencing reads mapped to each genomic position and provides important information for detecting genomic abnormalities such as duplications or deletions.

Coverage values were summarized across genomic windows to facilitate visualization and downstream statistical analysis.

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



2.3.6 Copy Number Variation Detection

Copy Number Variations (CNVs) were inferred using a coverage-based approach, where deviations from the expected sequencing depth were interpreted as potential duplications or deletions.

The CNV ratio was calculated by normalizing coverage values relative to the median genome coverage. Regions presenting significant deviations from the baseline coverage were classified as potential structural variants:

CNV ratio $< 1 \rightarrow$ potential deletions
CNV ratio $> 1 \rightarrow$ potential duplications

The initial CNV profile was generated using all annotated genomic features.
The distribution revealed that most regions had zero read coverage, producing a sparse CNV profile with extreme values.

This behavior is expected when low-coverage regions or non-coding features are included in the analysis because a large proportion of features had zero read counts, the median coverage was strongly affected, which distorted the CNV ratio calculation.

To improve the reliability of the CNV analysis, features with zero coverage were removed before recalculating the CNV ratio.

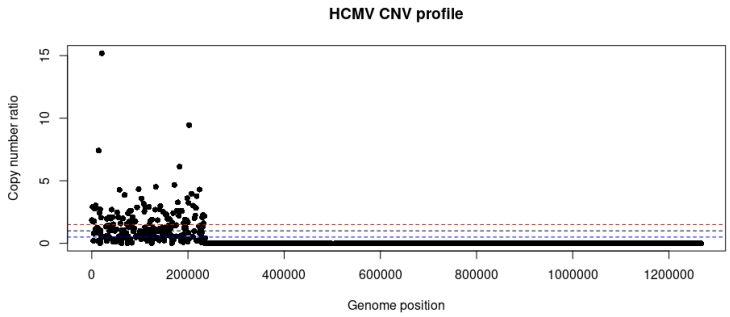
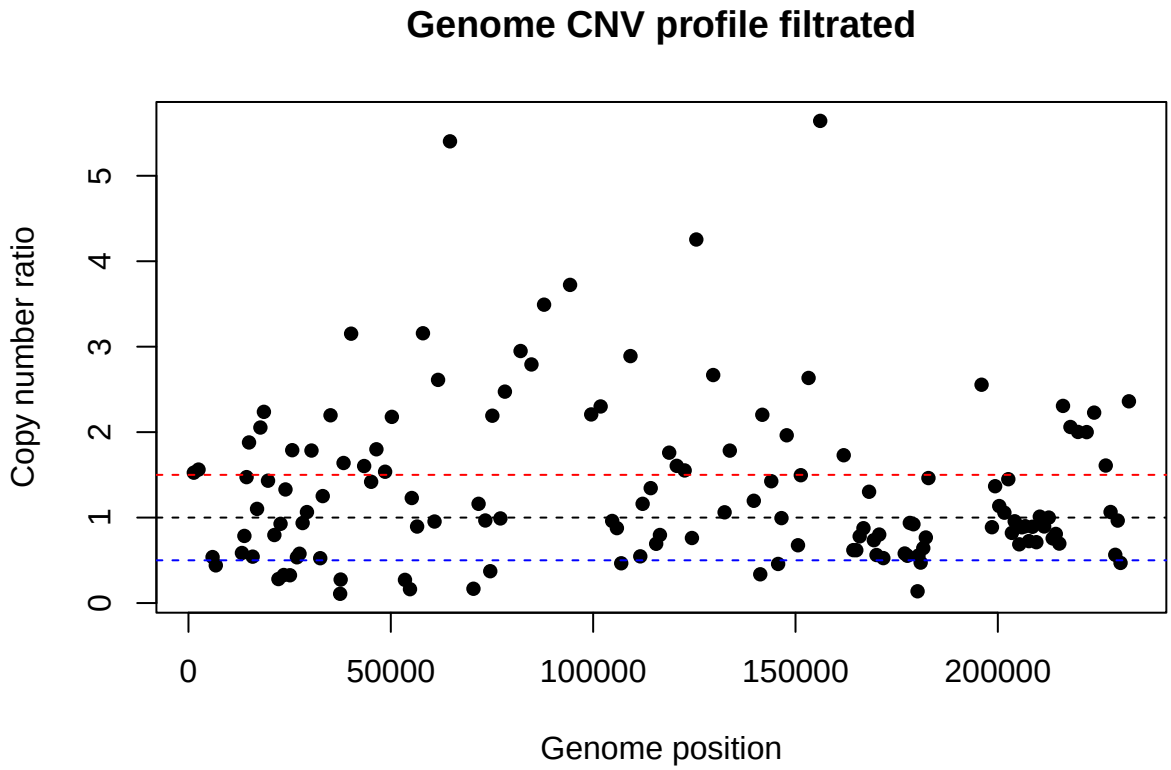


Figure 1: Distribuição de deleções e duplicações detectadas

Copy Number Variation Filtered
This filtering step produced a more stable coverage distribution and allowed the identification of potential duplication or deletion regions.



Regions with CNV ratio greater than 1.5 were considered potential duplications, while regions below 0.5 were considered potential deletions.

This approach allows identification of structural variation patterns along the viral genome.

3 Visualization and Genomic Annotation

All genomic analyses were conducted in R, using packages from the Bioconductor ecosystem, including:
- *GenomicRanges for genomic interval manipulation*
- *rtracklayer for importing genome annotations*
- *Gviz for genomic visualization*
- *ggplot2 for graphical representation of coverage patterns*

Genome annotation data were imported from the reference genome GenBank/GFF file, allowing mapping of detected CNV regions to specific viral genes.

Genomic plots were generated to visualize:
- *genome coverage distribution*
- *regions with potential duplications*
- *regions with potential deletions*

These visualizations facilitate interpretation of structural genomic variation across the HCMV genome.

4 Results

The most affected genes included RL1, UL55 and UL83, which accumulated the highest number of detected variants in the analyzed dataset.

4.1 Variant Distribution Across the Genome

A total of several genomic variants were detected across the HCMV genome after variant calling and filtering. These variants were mapped to annotated genomic regions using overlap analysis between variant coordinates and gene annotations.

The distribution of variants across the genome revealed heterogeneous mutation patterns, with certain genomic regions accumulating higher numbers of variants. Visualization of variant density along the genome indicated that mutations were not uniformly distributed, suggesting localized regions of higher genomic variability.

The coverage depth analysis revealed heterogeneous read distribution across the viral genome. While most genomic regions displayed coverage values consistent with the expected baseline, some

segments showed significant deviations in normalized coverage values.

These deviations suggest the presence of potential copy number variations, including candidate deletion and duplication regions. Further interpretation of these regions was performed by integrating coverage data with genome annotation.

4.2 Genes with Highest Mutation Frequency

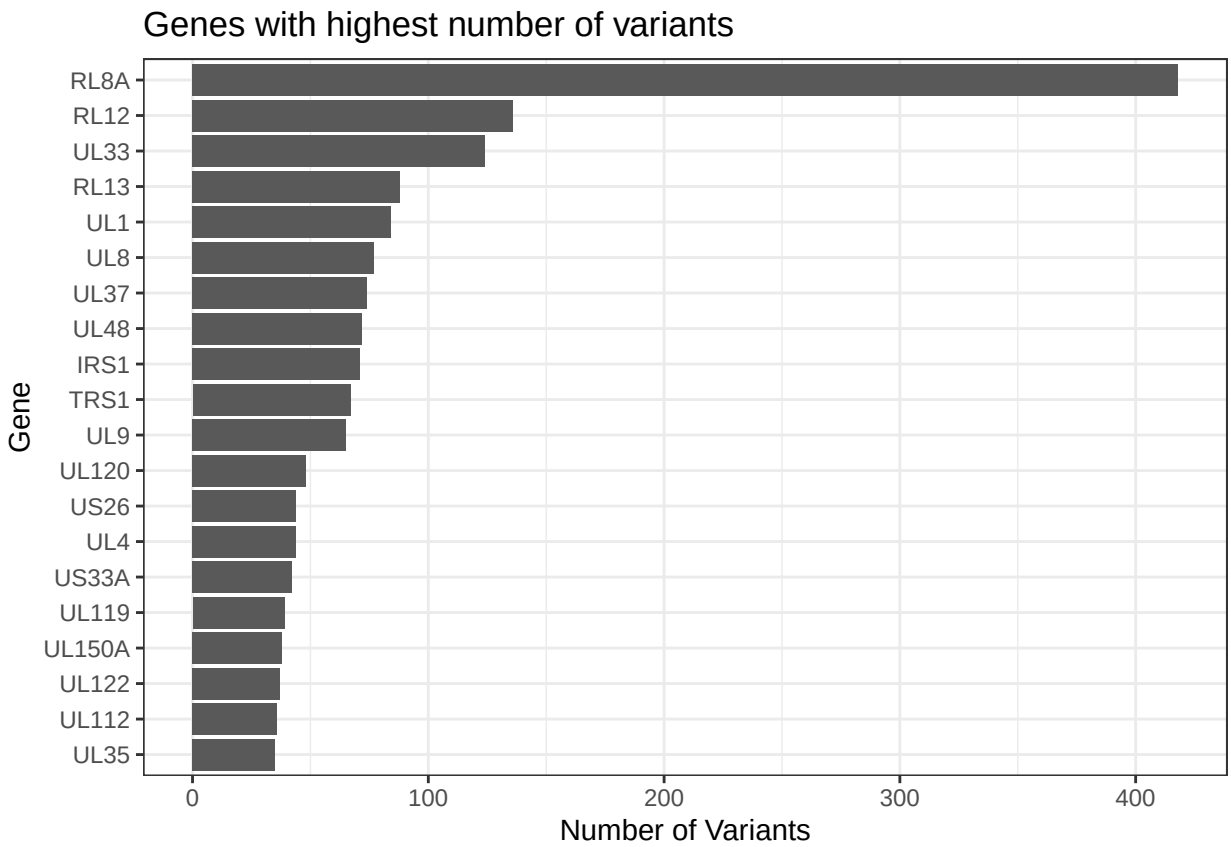
Variants located within annotated genes were summarized to determine which genes presented the highest number of mutations. The analysis revealed that a subset of viral genes accumulated a larger number of variants compared to the rest of the genome.

The bar plot below illustrates the genes presenting the highest number of detected variants. These genes may represent regions of increased evolutionary plasticity or may be subject to selective pressures during viral replication in the host.

A total of 2044 variants overlapped coding regions of the genome. Variant distribution across genes was heterogeneous, with several genes presenting higher mutation loads.

The genes presenting the highest number of variants are shown in the figure above and represent potential hotspots of genomic variability within the HCMV genome.

gene	n
RL8A	418
RL12	136
UL33	124
RL13	88
UL1	84
UL8	77
UL37	74
UL48	72
IRS1	71
TRS1	67



4.3 Mutation Density Normalized by Gene Length

To account for differences in gene size, mutation density was calculated by normalizing the number of detected variants by the length of each gene. This approach allows a more accurate comparison of mutation rates between genes of different sizes.

The analysis revealed that some genes with moderate numbers of variants exhibited relatively high mutation density due to their smaller genomic length.

4.4 Genome Coverage Depth

Genome coverage depth was calculated to evaluate the distribution of sequencing reads across the viral genome. Coverage analysis revealed substantial variability across genomic regions, reflecting differences in sequencing depth and potential biases associated with genomic structure.

The coverage profile showed that several genomic regions presented low or absent coverage, which may reflect sequencing limitations, alignment challenges in repetitive regions, or biological variability in viral genome representation.

4.5 Copy Number Variation Analysis

Copy number variation (CNV) analysis was performed using a coverage-based normalization approach. Coverage values for each genomic feature were normalized relative to the median genome coverage to infer relative copy number changes.

Initial analysis revealed a large number of features with zero coverage, which resulted in extreme CNV ratio values. After removing features with zero coverage, a more stable CNV profile was obtained.

The filtered CNV analysis identified genomic regions with coverage deviations relative to the baseline, suggesting potential duplication or deletion events. These structural variations may reflect genomic plasticity and adaptation mechanisms within the viral population.

5 Discussion

The genomic analysis performed in this study highlights the complexity of the HCMV genome and its capacity for genetic variation. The uneven distribution of variants across the genome suggests that certain viral genes may tolerate higher mutation rates or may be under different selective pressures during infection.

Regions presenting increased mutation frequency may play roles in viral adaptation to the host environment or immune evasion mechanisms. Additionally, coverage-based CNV analysis revealed potential structural variability in the viral genome, which has been previously reported in herpesviruses and may contribute to viral evolution.

However, coverage variability may also be influenced by sequencing and alignment biases, particularly in repetitive genomic regions characteristic of herpesvirus genomes. Therefore, further analyses combining long-read sequencing and structural variant detection tools could provide more detailed insights into these genomic rearrangements.

Overall, the integration of variant detection, coverage analysis, and CNV inference provides a comprehensive framework for exploring viral genomic diversity using next-generation sequencing data.

6 Conclusion

This study demonstrates the application of bioinformatic tools to investigate genomic variation in human cytomegalovirus using next-generation sequencing data. Variant analysis revealed heterogeneous mutation patterns across the viral genome, while coverage-based CNV analysis suggested the presence of potential structural variations.

These results highlight the importance of genomic approaches for understanding viral diversity and the structural dynamics of viral genomes. Such analyses may contribute to improved knowledge of viral evolution and may support future studies investigating viral adaptation and pathogenicity.